Physics 438B – Chapter #13
Time-Independent Perturbation Theory

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Introduction

There are relatively few Hamiltonians for which the energy eigenvalue problem

$$H|\psi\rangle = E|\psi\rangle$$

can be solved exactly. In many physically important situations, however, the Hamiltonian is close to another Hamiltonian whose eigenvalue problem we already know how to solve. The Hamiltonian we can solve exactly is called the unperturbed Hamiltonian, and the left-over portion is the so-called perturbation. We assume the perturbation is small so that we can use approximations to find corrections to the exact solutions. Our goal in this section is to discover how energies and energy eigenstates of the unperturbed Hamiltonian are affected by the perturbation. Before we develop the full machinery of time-independent perturbation theory, we consider a simple exactly solvable problem. We'll solve it and then expand the solutions in small parameters. This will give us some intuition for perturbation theory and provide a useful point of comparison for the general formalism.

1 A Spin-1/2 Example

Ignoring orbital degrees of freedom (measurements of $\mathbf{r}, \mathbf{p}$, etc.), a particle with spin will precess in a magnetic field. The Hamiltonian for a spin-1/2 system with an applied magnetic field in the z-direction, $\mathbf{B} = B_0\hat{\mathbf{z}}$, can be written as

$$H_0 = -\boldsymbol{\mu}\cdot\mathbf{B} = \omega_0 S_z = \frac{\hbar}{2}\begin{pmatrix}\omega_0 & 0 \\ 0 & -\omega_0\end{pmatrix}$$

where ω_0 is the Larmor frequency, proportional to the strength of the applied magnetic field B_0, and the matrix representation of H_0 is written in the ordered basis $\{|+z\rangle, |-z\rangle\}$. The energy eigenstates are the spin-up and spin-down states $|\pm z\rangle$ and the corresponding energy eigenvalues are

$$E_\pm^{(0)} = \pm\frac{\hbar\omega_0}{2},$$

where the superscript zero on the energy indicates the order of the solution—these are the unperturbed energies.

We perturb the system by changing the magnetic field. Since any change can be decomposed into a component along the original field direction and a component perpendicular, we write the new total field as $\mathbf{B} = B_0\hat{\mathbf{z}} + B_1\hat{\mathbf{z}} + B_2\hat{\mathbf{x}}$, and the Hamiltonian becomes

$$H = \omega_0 S_z + \omega_1 S_z + \omega_2 S_x = \frac{\hbar}{2}\begin{pmatrix} \omega_0+\omega_1 & \omega_2 \\ \omega_2 & -\omega_0-\omega_1 \end{pmatrix},$$

where ω_1 is proportional to B_1 and ω_2 is proportional to B_2. We have already solved the eigenvalue problem $H|\psi\rangle = E|\psi\rangle$ for this Hamiltonian in Chapter 3, so I will simply write the results below:

$$H|\pm n\rangle = E_\pm|\pm n\rangle,$$

where $|\pm n\rangle$ are the eigenstates of $S_n = \hat{\mathbf{n}}\cdot\mathbf{S}$ and $E_\pm$ are the corresponding eigenvalues. In this case, the direction $\hat{\mathbf{n}}$ of the magnetic field is given by the polar angle $\theta = \tan^{-1}\frac{\omega_2}{(\omega_0+\omega_1)}$ and the azimuthal angle $\phi = 0$ on the x-z plane. Explicitly, we have the energy eigenstates

$$|{+n}\rangle = \cos\frac{\theta}{2}|{+z}\rangle + \sin\frac{\theta}{2}|{-z}\rangle$$

$$|{-n}\rangle = -\sin\frac{\theta}{2}|{+z}\rangle + \cos\frac{\theta}{2}|{-z}\rangle$$

and the energies are

$$E_\pm = \pm\frac{\hbar\omega_0}{2}\sqrt{\left(1+\frac{\omega_1}{\omega_0}\right)^2 + \left(\frac{\omega_2}{\omega_0}\right)^2}.$$

Given the complete solution, we can now consider the perturbing field to be weak in the sense that $\omega_1 \ll \omega_0$ and $\omega_2 \ll \omega_0$. Notice that when $\omega_1 = \omega_2 = 0$, the problem reduces to that of spin precession about the z direction, i.e.,

$$|\pm n\rangle \to |\pm z\rangle, \quad E_\pm \to \pm\frac{\hbar\omega_0}{2}.$$

Let's start by expanding the energies to second order in ω_1/ω_0 and ω_2/ω_0:

$$\begin{aligned} E_\pm &= \pm\frac{\hbar\omega_0}{2}\sqrt{1+\frac{2\omega_1}{\omega_0}+\frac{\omega_1^2}{\omega_0^2}+\frac{\omega_2^2}{\omega_0^2}} \\ &\approx \pm\frac{\hbar\omega_0}{2}\left[1+\frac{1}{2}\left(\frac{2\omega_1}{\omega_0}+\frac{\omega_1^2}{\omega_0^2}+\frac{\omega_2^2}{\omega_0^2}\right)-\frac{1}{8}\left(\frac{2\omega_1}{\omega_0}+\frac{\omega_1^2}{\omega_0^2}+\frac{\omega_2^2}{\omega_0^2}\right)^2\right] \end{aligned}$$

$$= \pm\frac{\hbar\omega_0}{2}\left[1+\frac{\omega_1}{\omega_0}+\frac{\omega_2^2}{2\omega_0^2}\right] = \pm\frac{\hbar\omega_0}{2}\pm\frac{\hbar\omega_1}{2}\pm\frac{\hbar\omega_2^2}{4\omega_0}$$

Observe that the first "correction" to a given energy is linear or first order in the perturbation and equal to the *diagonal term* of the perturbation Hamiltonian, and the second correction is quadratic in the perturbation and proportional to the square of the *off-diagonal term* in the perturbation Hamiltonian. This general pattern of corrections is characteristic of perturbation theory.

As for the energy eigenstates, note that to first order,

$$\sin\frac{\theta}{2}\approx\frac{\omega_2}{2\omega_0}, \quad \text{and} \quad \cos\frac{\theta}{2}\approx 1.$$

Hence,

$$|{+n}\rangle \approx |{+z}\rangle + \frac{\omega_2}{2\omega_0}|{-z}\rangle$$

$$|{-n}\rangle = |{-z}\rangle - \frac{\omega_2}{2\omega_0}|{+z}\rangle$$

We conclude that the first-order eigenstate correction depends only on the off-diagonal matrix element, not on the diagonal elements. We can easily verify that the eigenstates are normalized to first order in the perturbation, e.g.,

$$\langle{+n}|{+n}\rangle = 1+\left(\frac{\omega_2}{2\omega_0}\right)^2 \approx 1.$$

2 Nondegenerate Perturbation Theory

Suppose we know how to solve the eigenvalue problem for a Hamiltonian H_0, written as

$$H_0|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle, \tag{1}$$

where the eigenstates form a complete orthonormal basis in the sense that

$$\langle m^{(0)}|n^{(0)}\rangle = \delta_{nm}, \quad \text{and} \quad \sum_m |m^{(0)}\rangle\langle m^{(0)}| = \mathbb{1}.$$

Technically, the expressions above are only relevant to the discrete portion of the Hamiltonian's spectrum. For the continuous portion, when relevant, it's understood that the Kronecker delta is replaced by a Dirac delta, and the sum over states becomes an integral.

We will stick to the discrete notation in these notes.

In perturbation theory, we consider adding a perturbation Hamiltonian V such that

$$H = H_0 + V, \tag{2}$$

and the goal is to approximate the eigenvalues and eigenstates of H using the known eigenvalues and eigenstates of H_0. It's useful to introduce a parameter λ and consider the family

$$H = H_0 + \lambda V. \tag{3}$$

The parameter λ allows us to interpolate between the unperturbed Hamiltonian ($\lambda = 0$) and the complete Hamiltonian containing the perturbation ($\lambda = 1$).

Assume that $E_n^{(0)}$ is a nondegenerate eigenvalue of H_0. As we increase λ, the unperturbed eigenvalue will change smoothly to the value E, and the corresponding eigenstate becomes $|\psi\rangle$. When $\lambda = 0$, $E = E_n^{(0)}$ and $|\psi\rangle = |n^{(0)}\rangle$. For $\lambda \neq 0$, the exact eigenvalue equation is

$$(H_0 + \lambda V)|\psi\rangle = E|\psi\rangle. \tag{4}$$

Let P denote the projector onto the unperturbed state $|n^{(0)}\rangle$, i.e., $P = |n^{(0)}\rangle\langle n^{(0)}|$, and let Q be its orthogonal complement, which simply means $Q = \mathbb{1} - P$. Recall that any vector in $\mathbb{R}^3$ may be decomposed into one component that is parallel to some unit vector $\hat{\mathbf{n}}$ and another component which is perpendicular. This decomposition works even in abstract vector spaces. For example, since $P + Q = \mathbb{1}$, we may write

$$|\psi\rangle = P|\psi\rangle + Q|\psi\rangle. \tag{5}$$

The state $|\psi\rangle$ is only defined up to an overall normalization factor, and it will be convenient to normalize according to the condition

$$\langle n^{(0)}|\psi\rangle = 1 \tag{6}$$

This is called "intermediate normalization," and it has the following wonderful consequence:

$$P|\psi\rangle = |n^{(0)}\rangle\langle n^{(0)}|\psi\rangle = |n^{(0)}\rangle \tag{7}$$

and therefore,

$$|\psi\rangle = |n^{(0)}\rangle + Q|\psi\rangle. \tag{8}$$

Thus, at least formally, $Q|\psi\rangle$ constitutes a "correction" to the original state. Because

$$Q = \sum_{m\neq n} |m^{(0)}\rangle\langle m^{(0)}|,$$

the correction consists entirely of states other than $|n^{(0)}\rangle$.

Since we already know the component of $|\psi\rangle$ along $|n^{(0)}\rangle$, we only need to solve for the remaining component $Q|\psi\rangle$. From the eigenvalue equation, it follows that

$$Q(E - H_0)|\psi\rangle = \lambda QV|\psi\rangle \tag{9}$$

Since Q is constructed from eigenvectors of H_0, it commutes with H_0, and we find

$$(E - H_0)Q|\psi\rangle = \lambda QV|\psi\rangle \tag{10}$$

Note that the operator $E - H_0$ is invertible as long as E does not lie within the spectrum of H_0 (it's not one of the eigenvalues of H_0). This is obviously a problem when $\lambda = 0$, since $E = E_n^{(0)}$. We can get around this by simply restricting the inverse to the subspace spanned by all eigenstates except for $|n^{(0)}\rangle$. In other words, define the reduced inverse operator

$$R = \sum_{m\neq n} \frac{|m^{(0)}\rangle\langle m^{(0)}|}{E - E_m^{(0)}}. \tag{11}$$

The inverse operator is explained a bit further in Appendix A. As long as $E \neq E_m^{(0)}$ for $m \neq n$, the operator R exists. From orthonormality of the eigenstates, it follows that

$$R(E - H_0)Q = Q, \quad \text{and} \quad RQ = R. \tag{12}$$

The expressions above may or may not be immediately obvious to you. When in doubt, write them out! Start by expanding R and Q in terms of the (unperturbed) energy eigenstates. If apply R from the left to both sides of the relation $Q(E - H_0)|\psi\rangle = \lambda QV|\psi\rangle$, we find

$$Q|\psi\rangle = \lambda RV|\psi\rangle, \tag{13}$$

and as a consequence,

$$\boxed{|\psi\rangle = |n^{(0)}\rangle + \lambda RV|\psi\rangle.} \tag{14}$$

This expression allows us to assess the correction to the unperturbed state in more detail. The size of the correction is ultimately controlled by the strength with which V couples different unperturbed states compared with their energy separation. For a weak perturbation, where $E \approx E_n^{(0)}$ and $|\psi\rangle \approx |n^{(0)}\rangle$, the coefficients are approximately

$$\frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}$$

when $\lambda = 1$. Perturbation theory is therefore expected to work well when these dimensionless ratios are small.

We'll return to our implicit expression for $|\psi\rangle$ once we've made progress on determining the energy E. Taking the inner product of the eigenvalue equation with the state $|n^{(0)}\rangle$, we find

$$\langle n^{(0)}|(H_0 + \lambda V)|\psi\rangle = E\langle n^{(0)}|\psi\rangle.$$

Since $H|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle$ and $\langle n^{(0)}|\psi\rangle = 1$, it follows that

$$\boxed{E = E_n^{(0)} + \lambda\langle n^{(0)}|V|\psi\rangle} \tag{15}$$

It should be emphasized at this stage that the expressions for E and $|\psi\rangle$ are exact.

First-Order Approximation

Eq. (14) is an implicit expression for $|\psi\rangle$ since it appears on both the left- and right-hand sides. We can substitute the expression for $|\psi\rangle$ into the right-hand side to find

$$|\psi\rangle = |n^{(0)}\rangle + \lambda RV\Big(|n^{(0)}\rangle + \lambda RV|\psi\rangle\Big) = |n^{(0)}\rangle + \lambda RV|n^{(0)}\rangle + \lambda^2 RVRV|\psi\rangle. \tag{16}$$

If we neglect the final term, which is second order in the perturbation, we obtain an expression for $|\psi\rangle$ in terms of $|n^{(0)}\rangle$ that is valid to first order. If we substitute the first-order approximation for $|\psi\rangle$ into the expression for E and neglect terms that are second order and higher, we arrive at a first-order approximation for the energy:

$$\boxed{E^{[1]} = E_n^{(0)} + \lambda\langle n^{(0)}|V|n^{(0)}\rangle} \tag{17}$$

The first-order approximation to the energy eigenstate is then given by

$$\boxed{|\psi^{[1]}\rangle = |n^{(0)}\rangle + \lambda \sum_{m\neq n} \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}|m^{(0)}\rangle} \tag{18}$$

since

$$R(E^{[1]}) = \sum_{m\neq n} \frac{|m^{(0)}\rangle\langle m^{(0)}|}{E_n^{(0)} + \lambda\langle n^{(0)}|V|n^{(0)}\rangle - E_m^{(0)}} = \sum_{m\neq n} \frac{|m^{(0)}\rangle\langle m^{(0)}|}{E_n^{(0)} - E_m^{(0)}} + O(\lambda). \tag{19}$$

It is common to refer to the correction terms themselves, so we denote the first-order correction to the energy $|E_n^{(1)}\rangle$ and the first-order correction to the energy eigenstate $|n^{(1)}\rangle$. Thus,

$$E^{[1]} = E_n^{(0)} + \lambda E_n^{(1)}, \quad \text{and} \quad |\psi^{[1]}\rangle = |n^{(0)}\rangle + \lambda|n^{(1)}\rangle \tag{20}$$

where

$$E_n^{(1)} = \langle n^{(0)}|V|n^{(0)}\rangle, \quad \text{and} \quad |n^{(1)}\rangle = \sum_{m\neq n} \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}|m^{(0)}\rangle \tag{21}$$

As you can see, the first-order correction to the energy is a diagonal matrix element of the perturbation Hamiltonian and the first-order correction to the state is a linear combination of the off-diagonal matrix elements of V and all states orthogonal to the unperturbed state.

Second-Order Approximation

Beginning with the exact expression for the energy, Eq. (15), and successively substituting the implicit expression for $|\psi\rangle$, we find

$$\begin{aligned}
E &= E_n^{(0)} + \lambda\langle n^{(0)}|V\Big(|n^{(0)}\rangle + \lambda RV|\psi\rangle\Big) \\
&= E_n^{(0)} + \lambda\langle n^{(0)}|V|n^{(0)}\rangle + \lambda^2\langle n^{(0)}|VRV\Big(|n^{(0)}\rangle + \lambda RV|\psi\rangle\Big) \\
&= E_n^{(0)} + \lambda\langle n^{(0)}|V|n^{(0)}\rangle + \lambda^2\langle n^{(0)}|VRV|n^{(0)}\rangle + O(\lambda^3)
\end{aligned}$$

Thus, to second order,

$$E^{[2]} = E_n^{(0)} + \lambda\langle n^{(0)}|V|n^{(0)}\rangle + \lambda^2\langle n^{(0)}|VRV|n^{(0)}\rangle \tag{22}$$

Since the final term is already proportional to λ^2, $R(E)$ need only be evaluated using the zeroth-order energy $E_n^{(0)}$ in this term. Using the definition of R then gives

$$\boxed{E_n^{(2)} = \langle n^{(0)}|VRV|n^{(0)}\rangle = \sum_{m\neq n} \frac{|\langle m^{(0)}|V|n^{(0)}\rangle|^2}{E_n^{(0)} - E_m^{(0)}}} \tag{23}$$

To obtain the state through second order, we similarly continue the successive substitution in the exact equation for $|\psi\rangle$:

$$|\psi\rangle = |n^{(0)}\rangle + \lambda RV|n^{(0)}\rangle + \lambda^2 RVRV\Big(|n^{(0)}\rangle + \lambda RV|\psi\rangle\Big)$$

$$= |n^{(0)}\rangle + \lambda RV|n^{(0)}\rangle + \lambda^2 RVRV|n^{(0)}\rangle + O(\lambda^3)$$

There is an additional subtlety in obtaining the second-order correction to the state since R depends on the energy E. Since the first term containing R is multiplied by λ, we must know $R(E)$ through first order in λ in order to determine $|\psi\rangle$ through second order. Using the second-order approximation to the energy, we have

$$R(E^{[2]}) = \sum_{m\neq n} \frac{|m^{(0)}\rangle\langle m^{(0)}|}{E_n^{(0)} - E_m^{(0)}} - \lambda\langle n^{(0)}|V|n^{(0)}\rangle \sum_{m\neq n} \frac{|m^{(0)}\rangle\langle m^{(0)}|}{\Big(E_n^{(0)} - E_m^{(0)}\Big)^2} + O(\lambda^2) \tag{24}$$

Although the explicit calculation is not shown, this follows from the series expansion

$$\frac{1}{a + \lambda b + \lambda^2 c} = \frac{1}{a} - \lambda\frac{b}{a^2} + O(\lambda^2)$$

Notice that the λ^2 correction to the energy does not contribute to R at the order required. Substituting this expansion for R and collecting all terms at each order of λ gives the second-order correction to the energy eigenstate:

$$\boxed{\begin{aligned} |n^{(2)}\rangle = & \sum_{m\neq n}\sum_{k\neq n} \frac{\langle m^{(0)}|V|k^{(0)}\rangle\langle k^{(0)}|V|n^{(0)}\rangle}{\Big(E_n^{(0)} - E_m^{(0)}\Big)\Big(E_n^{(0)} - E_k^{(0)}\Big)}|m^{(0)}\rangle \\ & - \sum_{m\neq n} \frac{\langle n^{(0)}|V|n^{(0)}\rangle\langle m^{(0)}|V|n^{(0)}\rangle}{\Big(E_n^{(0)} - E_m^{(0)}\Big)^2}|m^{(0)}\rangle \end{aligned}} \tag{25}$$

Summary of Key Results

Define

$$V_{mn} \equiv \langle m^{(0)}|V|n^{(0)}\rangle, \quad \text{and} \quad \Delta_{nm} \equiv E_n^{(0)} - E_m^{(0)}. \tag{26}$$

The second-order approximation to the energy is

$$E^{[2]} = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} \tag{27}$$

and the second-order approximation to the state is

$$|\psi^{[2]}\rangle = |n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle \tag{28}$$

The first- and second-order corrections to the energy are

$$E_n^{(1)} = V_{nn} \tag{29}$$

and

$$E_n^{(2)} = \sum_{m\neq n} \frac{|V_{mn}|^2}{\Delta_{nm}} \tag{30}$$

The corresponding corrections to the state are

$$|n^{(1)}\rangle = \sum_{m\neq n} \frac{V_{mn}}{\Delta_{nm}}|m^{(0)}\rangle \tag{31}$$

and

$$|n^{(2)}\rangle = \sum_{m\neq n}\sum_{k\neq n} \frac{V_{mk}V_{kn}}{\Delta_{nm}\Delta_{nk}}|m^{(0)}\rangle - \sum_{m\neq n} \frac{V_{nn}V_{mn}}{\Delta_{nm}^2}|m^{(0)}\rangle \tag{32}$$

These results have been obtained using intermediate normalization, $\langle n^{(0)}|\psi\rangle = 1$. Consequently, all corrections to the state are orthogonal to $|n^{(0)}\rangle$. Normalization of the complete approximate state may be imposed afterward if needed.

Setting $\lambda = 1$

We introduced $H = H_0 + \lambda V$ to construct a perturbative solution. The physical Hamiltonian is $H = H_0 + V$, which corresponds to $\lambda = 1$. Therefore, after constructing the approximation to the desired order, we set $\lambda = 1$. For example, the energy through second order is

$$E_n \approx E_n^{(0)} + V_{nn} + \sum_{m\neq n} \frac{|V_{mn}|^2}{\Delta_{nm}}$$

Setting $\lambda = 1$ does not mean we are assuming 1 is a small number. The actual measure of the strength of the perturbation appears in dimensionless quantities such as $|V_{mn}/\Delta_{nm}|$ Thus, λ serves primarily as a bookkeeping parameter that organizes the successive corrections. The physical smallness of the perturbation is determined by the matrix elements of V relative to the relevant unperturbed energy gaps.

Example 2.1: Harmonic Oscillator with a Linear Perturbation

Consider a one-dimensional harmonic oscillator with mass m and angular frequency ω. The unperturbed Hamiltonian for this system is

$$H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

A weak linear potential is added as a perturbation, namely

$$V(x) = f_0 x,$$

where f_0 is a small positive constant. The full Hamiltonian is $H = H_0 + V$.

(a) Solve the full Hamiltonian by completing the square. In other words, determine the exact solutions to the energy eigenvalue equation.

(b) Expand the exact energies and eigenstates for small f_0.

(c) Use first-order perturbation theory to find the energy corrections,

(d) and then find the corrections to the energy eigenstates.

Solution. **(a)** The Hamiltonian is quadratic in x even with a linear term. Completing the square in x, we find

$$\begin{aligned}\frac{1}{2}m\omega^2 x^2 + \lambda f_0 x &= \frac{m\omega^2}{2}\left(x^2 + \frac{2\lambda f_0}{m\omega^2}x\right)\\ &= \frac{m\omega^2}{2}\left(x + \frac{\lambda f_0}{m\omega^2}\right)^2 - \frac{\lambda^2 f_0^2}{2m\omega^2}\end{aligned}$$

This potential describes a harmonic oscillator with a shifted equilibrium position

$$x_0 = -\frac{\lambda f_0}{m\omega^2}$$

and a constant energy shift. Thus, the exact energy eigenvalues are

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{\lambda^2 f_0^2}{2m\omega^2}$$

The exact eigenstates, in position space, are the harmonic oscillator eigenstates centered at the displaced equilibrium position:

$$\psi_n(x) = \psi_n^{(0)}\left(x + \frac{\lambda f_0}{m\omega^2}\right)$$

(b) Evidently, the leading order correction to the energy is second order:

$$E_n^{(1)} = 0$$
$$E_n^{(2)} = \frac{f_0^2}{2m\omega^2}$$

From the exact eigenfunctions, we can Taylor expand about $f_0 = 0$ to find the first-order correction to the state:

$$\psi_n(x) = \psi_n^{(0)}\left(x + \frac{\lambda f_0}{m\omega^2}\right) \approx \psi_n^{(0)} + \frac{\lambda f_0}{m\omega^2}\frac{d\psi_n^{(0)}}{dx}$$

Therefore, the first-order correction is

$$\psi_n^{(1)}(x) = \frac{f_0}{m\omega^2}\frac{d\psi_n^{(0)}}{dx}$$

It will be convenient to work with the energy eigenkets rather than the position-space wave functions. Using

$$p = -i\sqrt{\frac{m\hbar\omega}{2}}\left(a - a^\dagger\right)$$

and $d/dx = ip/\hbar$, we have

$$\frac{d}{dx}|n\rangle = \sqrt{\frac{m\hbar\omega}{2}}\left(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle\right)$$

Hence,

$$|n^{(1)}\rangle = \frac{f_0}{\sqrt{2m\hbar\omega^3}}\left(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle\right)$$

(c) According to first-order perturbation theory,

$$E_n^{(1)} = \langle n^{(0)}|V|n^{(0)}\rangle = f_0\langle n^{(0)}|x|n^{(0)}\rangle$$

The diagonal matrix elements are the expectation values of x in the energy eigenstates. Since the energy eigenstates are even or odd about the center of the potential ($x = 0$ for the unperturbed system), the diagonal elements vanish. Thus, $E_n^{(1)} = 0$.

(d) The first-order correction to the eigenstate is

$$|n^{(1)}\rangle = \sum_{m\neq n} \frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle.$$

Using

$$x = \sqrt{\frac{\hbar}{2m\omega}} \left(a + a^\dagger\right),$$

we find

$$x|n^{(0)}\rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle\right).$$

The only nonzero matrix elements are

$$\langle n-1|x|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n}$$

and

$$\langle n+1|x|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n+1}$$

and the corresponding energy gaps are

$$E_n^{(0)} - E_{n-1}^{(0)} = \hbar\omega, \quad \text{and} \quad E_n^{(0)} - E_{n+1}^{(0)} = -\hbar\omega$$

Therefore, first-order correction is

$$|n^{(1)}\rangle = \frac{f_0\sqrt{\frac{\hbar}{2m\omega}}\sqrt{n}}{\hbar\omega}|n-1\rangle - \frac{f_0\sqrt{\frac{\hbar}{2m\omega}}\sqrt{n+1}}{\hbar\omega}|n+1\rangle,$$

or

$$|n^{(1)}\rangle = \frac{f_0}{\sqrt{2m\hbar\omega^3}} \left(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle\right).$$

Example 2.2: Infinite Square Well with a Linear Perturbation

Consider a particle of mass m confined to an infinite square well extending from $x = 0$ to $x = L$. The unperturbed Hamiltonian is denoted H_0. A weak linear potential is added as a perturbation, namely

$$V(x) = f_0 x,$$

where f_0 is a small positive constant. The full Hamiltonian is $H = H_0 + V$.

(a) Find the first-order corrections to the energy eigenvalues $E_n^{(1)}$.

(b) Find the off-diagonal matrix elements $\langle m^{(0)}|V|n^{(0)}\rangle$, $m \neq n$.

(c) Determine the first-order corrections to the energy eigenstates $|n^{(1)}\rangle$.

(d) Identify a dimensionless measure of the strength of the perturbation and state a condition under which first-order perturbation theory should be reliable.

Solution. The unperturbed eigenstates and energies are

$$\psi_n^{(0)} = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n^{(0)} = \frac{n^2\pi^2\hbar^2}{2mL^2}.$$

(a) Formally, first-order correction to the energy eigenvalue is

$$E_n^{(1)} = \langle n^{(0)}|V|n^{(0)}\rangle = f_0\langle n^{(0)}|x|n^{(0)}\rangle$$

In position space,

$$E_n^{(1)} = \frac{2f_0}{L}\int_0^L x \sin^2\left(\frac{n\pi x}{L}\right) dx$$

Using $2\sin^2(\theta) = 1 - \cos 2\theta$,

$$E_n^{(1)} = \frac{f_0}{L}\int_0^L x\left[1 - \cos\left(\frac{2n\pi x}{L}\right)\right] dx = \boxed{\frac{f_0 L}{2}}$$

(b) We solved for the off-diagonal matrix elements of the position operator in Group Activity #1, but we'll re-derive the result here anyways. Letting $V_{mn} = \langle m^{(0)}|V|n^{(0)}\rangle$,

$$V_{mn} = \frac{2f_0}{L}\int_0^L x \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Using $2\sin A\sin B = \cos(A-B) - \cos(A+B)$,

$$V_{mn} = \frac{f_0}{L}\int_0^L x\left[\cos\left(\frac{(m-n)\pi x}{L}\right) - \cos\left(\frac{(m+n)\pi x}{L}\right)\right]dx.$$

Since

$$\frac{\partial}{\partial a}\sin(ax) = x\cos(ax),$$

it follows that

$$\int x\cos(ax)dx = \frac{d}{da}\int \sin(ax)dx = \frac{d}{da}\left(-\frac{\cos(ax)}{a}\right) = \frac{\cos(ax)}{a^2} + \frac{x\sin(ax)}{a}$$

In our case, $a = (m\pm n)\pi/L$, so the product ax evaluates to $(m\pm n)\pi$ or 0 under the definite integral, and in both cases $\sin(ax) = 0$. Hence,

$$V_{mn} = \frac{f_0}{L}\frac{L^2}{\pi^2}\left[\left(\frac{\cos((m-n)\pi)-1}{(m-n)^2}\right) - \left(\frac{\cos((m+n)\pi)-1}{(m+n)^2}\right)\right]$$

If $m+n$ is even, so is $m-n$, and $\cos((m\pm n)\pi) = 1$. We can see above that both terms vanish, so $V_{mn} = 0$ when $m+n$ is even. On the other hand, if $m+n$ is odd, so is $m-n$, and $\cos((m\pm n)\pi) = -1$. For $m+n$ odd, we find

$$V_{mn} = -\frac{2f_0L}{\pi^2}\left[\frac{1}{(m-n)^2} - \frac{1}{(m+n)^2}\right] = -\frac{8f_0L}{\pi^2}\frac{mn}{(m^2-n^2)^2}$$

Therefore,

$$\boxed{V_{mn} = \begin{cases} -\dfrac{8f_0L}{\pi^2}\dfrac{mn}{(m^2-n^2)^2}, & m+n \text{ odd},\\[2ex] 0, & m+n \text{ even}.\end{cases}}$$

It is interesting that the perturbation mixes only states with opposite parity.

(c) Formally, the first-order correction to the state is

$$|n^{(1)}\rangle = \sum_{m\neq n}\frac{\langle m^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}|m^{(0)}\rangle.$$

We've already found the off-diagonal matrix elements of the perturbation, so all we need are the energy gaps:

$$E_n^{(0)} - E_m^{(0)} = \frac{\pi^2\hbar^2}{2mL^2}\,(n^2 - m^2).$$

Thus, for $m+n$ odd,

$$\frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} = \frac{16mf_0L^3}{\pi^4\hbar^2}\,\frac{mn}{(m^2-n^2)^3}$$

and the first-order correction to the energy eigenstate is

$$|n^{(1)}\rangle = \frac{16mf_0L^3}{\pi^4\hbar^2}\sum_{\substack{m\neq n\\ m+n \text{ odd}}}\frac{mn}{(m^2-n^2)^3}|m^{(0)}\rangle.$$

In position space,

$$\boxed{\psi_n^{(1)}(x) = \frac{16mf_0L^3}{\pi^4\hbar^2}\sum_{\substack{m\neq n\\ m+n \text{ odd}}}\frac{mn}{(m^2-n^2)^3}\psi_m^{(0)}(x).}$$

(d) The coefficients in the first-order correction to the eigenstate contain the dimensionless quantity

$$\frac{mf_0L^3}{\hbar^2}$$

Therefore, first-order perturbation theory should be reliable when $f_0 \ll \hbar^2/mL^3$.

3 Degenerate Perturbation Theory

Now suppose the unperturbed energy $E_n^{(0)}$ is d_n-fold degenerate, i.e.,

$$H_0|n,a\rangle = E_n^{(0)}|n,a\rangle, \quad a = 1, \ldots, d_n. \tag{33}$$

Here n labels a distinct unperturbed energy eigenspace while a labels the different states within that space. The states satisfy

$$\langle n,a|m,b\rangle = \delta_{mn}\delta_{ab}$$

The projector onto the n-eigenspace associated with $E_n^{(0)}$ is

$$P_n = \sum_{a=1}^{d_n} |n,a\rangle\langle n,a| \tag{34}$$

while its orthogonal complement is

$$Q_n = \mathbb{1} - P_n = \sum_{m\neq n}\sum_{a=1}^{d_m} |m,a\rangle\langle m,a| \tag{35}$$

where d_m denotes the degeneracy of the mth eigenspace.

As before, we consider an exact eigenstate $|\psi\rangle$ satisfying $(H_0 + \lambda V)|\psi\rangle = E|\psi\rangle$, and decompose the state as

$$|\psi\rangle = P_n|\psi\rangle + Q_n|\psi\rangle \tag{36}$$

There are many states with the same unperturbed energy, so we do not yet know which linear combinations of them will correspond to the perturbed eigenstates. In the end, we may find multiple solutions with different perturbed energies, in which case we say that the degeneracy has been "lifted" by the perturbation.

The determination of $Q_n|\psi\rangle$ is mostly the same as the nondegenerate case, with the key difference being a new definition for the reduced inverse:

$$R_n = \sum_{m\neq n}\sum_{a=1}^{d_m} \frac{|m,a\rangle\langle m,a|}{E - E_m^{(0)}} \tag{37}$$

The new implicit equation for $|\psi\rangle$ is

$$|\psi\rangle = P_n|\psi\rangle + \lambda R_n V|\psi\rangle \tag{38}$$

This is almost identical to the expression obtained in the nondegenerate case. The difference is that $P_n|\psi\rangle$ is now an unknown vector within a multidimensional subspace. We therefore need another equation to determine $P_n|\psi\rangle$. Apply P_n to the eigenvalue equation:

$$P_n(H_0 + \lambda V)|\psi\rangle = EP_n|\psi\rangle \tag{39}$$

Since $P_nH_0 = E_n^{(0)}P_n$, we find

$$(E - E_n^{(0)})P_n|\psi\rangle = \lambda P_nV|\psi\rangle \tag{40}$$

Substituting Eq. (38) into the right-hand side of Eq. (40), we find

$$(E - E_n^{(0)})P_n|\psi\rangle = \lambda P_nVP_n|\psi\rangle + O(\lambda^2) \tag{41}$$

Since $P|\psi\rangle$ is a linear combination of the eigenstates within the degenerate subspace,

$$P|\psi\rangle = \sum_{a=1}^{d_n} c_a|n,a\rangle \tag{42}$$

where c_a are expansion coefficients. Taking the inner product with $|n,a\rangle$,

$$\langle n,a|P|\psi\rangle = \langle n,a|\psi\rangle = c_a. \tag{43}$$

We then take the inner product of Eq. (41) with $|n,a\rangle$ and ignore terms that are second order and higher in the perturbation. The first-order approximation to the energy is given implicitly by

$$\left(E^{[1]} - E_n^{(0)}\right)c_a = \lambda\sum_b \langle n,a|V|n,b\rangle c_b \tag{44}$$

Set $\lambda = 1$ to obtain the physical Hamiltonian and define

$$\Delta_n = E^{[1]} - E_n^{(0)}, \quad \text{and} \quad V_{ab} = \langle n,a|V|n,b\rangle \tag{45}$$

Eq. (44) then takes the form of an eigenvalue equation:

$$\sum_b V_{ab}c_b = \Delta_n c_a \tag{46}$$

Thus, finding the first-order corrections reduces to an ordinary $d_n \times d_n$ matrix eigenvalue problem. The eigenvalues of the perturbation matrix V_{ab} give the possible first-order shifts

Δ_n of the originally degenerate energy $E_n^{(0)}$, while the corresponding eigenvectors determine the particular linear combinations of the degenerate unperturbed states that become the perturbed eigenstates. If the eigenvalues of V_{ab} are distinct, the degeneracy is lifted at first order; if some remain equal, the degeneracy is only partially lifted (or not at all).

Once these linear combinations have been determined, the first-order correction to each state is found in essentially the same way as in the nondegenerate case. The only difference is that the sum over intermediate states must exclude the entire degenerate eigenspace associated with $E_n^{(0)}$, rather than a single state.

Example 3.1: A 3-Level System with Degeneracy

Consider a three-level quantum system with the following unperturbed Hamiltonian

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \Delta \end{pmatrix}, \quad \Delta > 0,$$

and the perturbation

$$V = \begin{pmatrix} 0 & g & a \\ g & 0 & b \\ a & b & 0 \end{pmatrix}$$

Assume that g, a, b are small compared to Δ. Observe that the first two unperturbed states are degenerate. First, determine the first-order energy corrections and the corresponding zeroth-order states selected by the perturbation within the degenerate subspace. Then determine the first-order corrections to these states. Lastly, compute the first-order corrections to the energy and state of the single nondegenerate eigenstate.

Solution. The unperturbed Hamiltonian is already diagonal in the standard orthonormal basis $\{|i\rangle\}$, i.e.,

$$H_0|1\rangle = 0, \quad H_0|2\rangle = 0, \quad H_0|3\rangle = \Delta|3\rangle$$

The zero-energy eigenspace is two-dimensional with basis $|1,1\rangle = |1\rangle$ and $|1,2\rangle = |2\rangle$ with $E_1^{(0)} = 0$. The remaining state belongs to a different eigenspace with $E_2^{(0)} = \Delta$

and $|2,1\rangle = |3\rangle$. The projector onto the degenerate eigenspace is

$$P_1 = |1\rangle\langle 1| + |2\rangle\langle 2| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and the perturbation restricted to this subspace is

$$V_1 = \begin{pmatrix} \langle 1|V|1\rangle & \langle 1|V|2\rangle \\ \langle 2|V|1\rangle & \langle 2|V|2\rangle \end{pmatrix} = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$$

We must now diagonalize this matrix to find the unperturbed states and first-order energy corrections. The characteristic equation is

$$\det \begin{pmatrix} -\Delta^{(1)} & g \\ g & -\Delta^{(1)} \end{pmatrix} = 0, \quad \text{or} \quad \left(\Delta^{(1)}\right)^2 - g^2 = 0$$

The corresponding energy shifts are

$$\Delta_+^{(1)} = g, \quad \text{and} \quad \Delta_+^{(1)} = -g.$$

For $\Delta_+^{(1)} = g$,

$$\begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = g \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

which gives $c_1 = c_2$. Choosing normalized coefficients, let

$$|1_+^{(0)}\rangle = \frac{1}{\sqrt{2}} \left(|1\rangle + |2\rangle\right)$$

be the first eigenstate to zeroth order. Similarly, for $\Delta_-^{(1)} = -g$, we find $c_1 = -c_2$, and therefore

$$|1_-^{(0)}\rangle = \frac{1}{\sqrt{2}} \left(|1\rangle - |2\rangle\right)$$

is the second eigenstate to zeroth order. Thus, the originally degenerate energy levels split at first order:

$$\boxed{E_+^{[1]} = g, \quad E_-^{[1]} = -g}$$

and the correct zeroth states are given by

$$\boxed{|1_+^{(0)}\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + |2\rangle\right), \quad |1_-^{(0)}\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle - |2\rangle\right)}$$

First-order corrections to the degenerate states. At this point, the degeneracy problem has been resolved within the original degenerate subspace. We can now use the same formula as in nondegenerate perturbation theory:

$$|1_\pm^{(1)}\rangle = \sum_{m\neq 1}\sum_b \frac{\langle m,b|V|1_\pm^{(0)}\rangle}{E_1^{(0)} - E_m^{(0)}}|m,b\rangle$$

where $|m,b\rangle$ denotes the zeroth-order eigenstates outside the degenerate subspace. In this case, there is only one such state, namely $|2,1\rangle = |3\rangle$. Thus,

$$|1_\pm^{(1)}\rangle = \frac{\langle 3|V|1_\pm^{(0)}\rangle}{0-\Delta}|3\rangle$$

For the state $|1_+^{(1)}\rangle$,

$$\langle 3|V|1_+^{(0)}\rangle = \frac{1}{\sqrt{2}}\left(\langle 3|V|1\rangle + \langle 3|V|2\rangle\right) = \frac{a+b}{\sqrt{2}}$$

Therefore,

$$\boxed{|1_+^{(1)}\rangle = -\frac{a+b}{\sqrt{2}\,\Delta}|3\rangle}$$

Similarly,

$$\langle 3|V|1_-^{(0)}\rangle = \frac{a-b}{\sqrt{2}}$$

and therefore,

$$\boxed{|1_-^{(1)}\rangle = -\frac{a-b}{\sqrt{2}\,\Delta}|3\rangle}$$

First-order corrections for the nondegenerate state. The first-order correction to the energy E_2 (the energy of the *second* eigenspace) is simply given by

$$\boxed{E_2^{(1)} = \langle 3|V|3\rangle = 0}$$

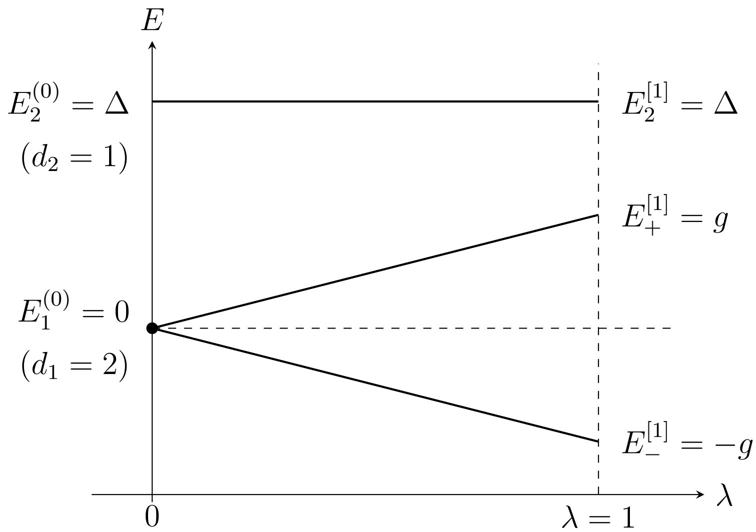
In other words, the energy does not change to first order in the perturbation. The first-order correction to $|2,1\rangle = |3\rangle$ is

$$|3^{(1)}\rangle = \frac{\langle 1_+^{(0)}|V|3\rangle}{\Delta}|1_+^{(0)}\rangle + \frac{\langle 1_-^{(0)}|V|3\rangle}{\Delta}|1_-^{(0)}\rangle$$

After working through some calculations, we find a rather simple result:

$$\boxed{|3^{(1)}\rangle = \frac{a}{\Delta}|1\rangle + \frac{b}{\Delta}|2\rangle}$$

See the figure below for a schematic representation of the energy levels as a function of λ, illustrating how the perturbation lifts the degeneracy at first order.



A Inverse Operators

Consider the operator $B = c\mathbb{1} - A$ where c is a scalar and the operator A is Hermitian. We are interested in finding the inverse operator, B^{-1}, such that $B^{-1}B = BB^{-1} = \mathbb{1}$, and possibly the conditions for its existence. On the eigenspace of A,

$$B|a_n\rangle = (c - a_n)|a_n\rangle.$$

Provided $c \neq a_n$, we can divide both sides by $c - a_n$ and write

$$B\left(\frac{1}{c - a_n}\right)|a_n\rangle = |a_n\rangle.$$

Furthermore, if c cannot be found anywhere within the spectrum of A, then the equation above holds for all n. Since A is Hermitian, it has a complete, orthonormal set of eigenvectors. Thus,

$$\mathbb{1} = \sum_n |a_n\rangle\langle a_n| = B\sum_n \frac{|a_n\rangle\langle a_n|}{c - a_n} = \sum_n \frac{|a_n\rangle\langle a_n|}{c - a_n}B,$$

and it follows that

$$B^{-1} = (c\mathbb{1} - A)^{-1} = \sum_n \frac{|a_n\rangle\langle a_n|}{c - a_n}$$